CHINA · TECHNOLOGY · AI FRONTIER

CHINESE ARTIFICIAL INTELLIGENCE

DeepSeek · Qwen · Agents
Architectures, efficiency and applications

Sergio Gevatschnaider



BLOCK 01 / 07

China's AI ecosystem

From research to deployment

Where does China compete, and where does it face constraints?

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AI competition has multiple dimensions

Technical capability, availability, infrastructure and adoption are distinct dimensions.

01

Model frontier

Reasoning, code, multimodality and robustness require different tests.

02

Open ecosystem

Accessible weights, tools, communities, deployment and specialised adaptation.

03

Infrastructure

Computing, memory, networks, data, energy, talent and software.

04

Adoption

Business integration, measured productivity, governance and safety.

AI's economic impact depends on both technology and diffusion.

Sources: Stanford AI Index 2026 · Official documentation

AI Index 2026: comparative evidence

Counting releases does not measure overall leadership.

59

Notable U.S. models identified in 2025

35

Notable models with Chinese affiliation in 2025

These figures do not establish equal quality, cost or performance across families.

2025 data in AI Index 2026. National affiliations may overlap.

Sources: Stanford AI Index 2026 · Official documentation

Mapping China's AI ecosystem

Firms, research, platforms and applications form a heterogeneous system.

01

DeepSeek

Efficient architectures and reasoning models.

02

Alibaba / Qwen

Open model families, cloud and multimodal agents.

03

Baidu / ERNIE

Models, search services and business ecosystems.

04

Tencent / Hunyuan

Applications, services and an AI platform.

Chinese firms have different strategies and capabilities.

Sources: Stanford AI Index 2026 · Official documentation

From laboratories to industry

AI creates value by solving measurable tasks.



Adoption requires process redesign as well as access to models.

Sources: Stanford AI Index 2026 · Official documentation

Strengths and constraints

Algorithmic efficiency can partially address infrastructure constraints.

A

STRENGTHS

Technical talent, manufacturing, demand, open software ecosystems, experimentation and rapid integration.

B

CONSTRAINTS

Advanced chips, memory, infrastructure, safety, data quality and access to markets.

Model capability ≠ hardware availability ≠ productivity.

Sources: Stanford AI Index 2026 · Official documentation

BLOCK 02 / 07

Model architectures

How models learn, process and generate

Where does the innovation behind the brands take place?

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From text to tokens

Tokenisation and embeddings turn language into numerical representations.



Tokens, words and parameters are different concepts.

Sources: Transformers · IBM MoE · Qwen and DeepSeek documentation

Transformers: attention and context

Attention links positions and builds representations that depend on context.

01

Query (Q)

What information does the current position seek?

02

Key (K)

How does each position identify relevant information?

03

Value (V)

What content enters the combined representation?

04

Masking

Prevents a decoder from attending to future tokens during prediction.

Attention cost depends on sequence length and architecture.

Sources: Transformers · IBM MoE · Qwen and DeepSeek documentation

Dense models and Mixture of Experts

Total capacity differs from the computation activated per token.

01

DENSE

Dense blocks apply their parameters to every token. Computation scales with the block size.

02

MoE

A router directs each token to a subset of experts, combining large total capacity with sparse activation.

MoE requires load balancing and communication across devices.

Sources: Transformers · IBM MoE · Qwen and DeepSeek documentation

How an MoE router works

Top-k routing assigns each token to selected computational paths.



Illustrative top-2 routing. Experts are learned sub-networks without guaranteed topic labels.

Sources: Transformers · IBM MoE · Qwen and DeepSeek documentation

Multimodality and action

A multimodal agent combines perception, language, tools and responses.

01

Image / video

Visual and temporal representations.

02

Text / code

Reasoning and symbolic generation.

03

Tools

APIs, files, browsing and restricted execution.

04

Action

Interfaces, automation and observable outcomes.

A multimodal model does not by itself constitute an autonomous agent.

Sources: Transformers · IBM MoE · Qwen and DeepSeek documentation

BLOCK 03 / 07

DeepSeek: efficiency and reasoning

Design matters alongside scale

What do V3 and R1 contribute, and what do their results leave unproven?

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DeepSeek-V3: sparse architecture

The original report documents a large MoE model with reduced activation.

671 B

Reported total parameters

37 B

Parameters activated per token

These figures do not establish equal quality, cost or performance across families.

DeepSeek-V3, December 2024 report. Do not extrapolate to later versions.

Sources: DeepSeek-V3 Technical Report · Nature (R1)

DeepSeek-V3: key innovations

Several architecture and training decisions contribute to efficiency.

01

MLA

Multi-head Latent Attention compresses key/value states.

02

DeepSeekMoE

Specialisation and selective expert activation.

03

Balancing

Methods for distributing token traffic.

04

MTP

A multi-token prediction training objective.

Evaluate the whole system alongside parameter counts.

Sources: DeepSeek-V3 Technical Report · Nature (R1)

Training cost requires a defined scope

Published GPU hours do not describe the complete cost of a project.

A

WHAT THE REPORT STATES

The V3 report describes 2.788 million H800 GPU hours for its documented training stages.

B

WHAT THIS FIGURE EXCLUDES

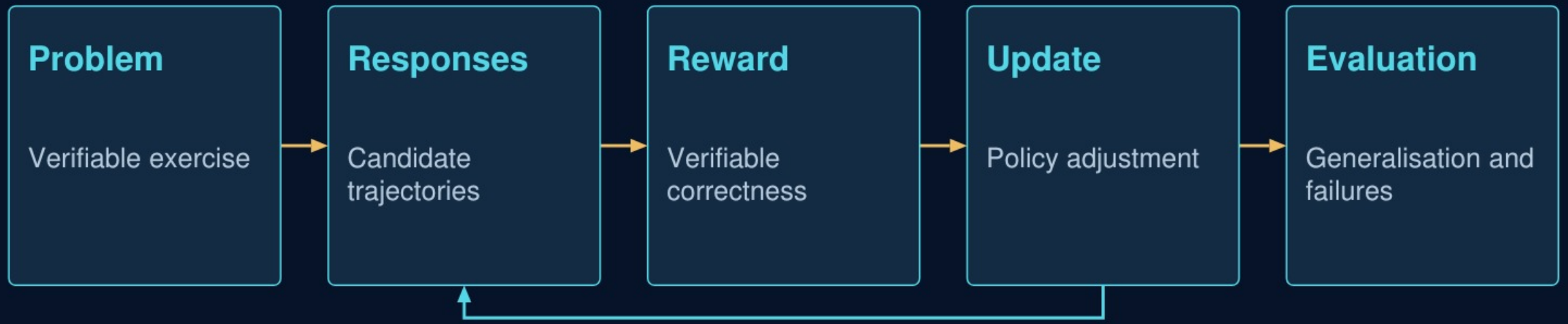
Earlier research, architecture exploration, data work, staff, infrastructure and other operating stages.

A training-cost estimate is different from a fully audited project cost.

Sources: DeepSeek-V3 Technical Report · Nature (R1)

DeepSeek-R1: reinforcement learning

Rewards on verifiable tasks can reinforce reasoning behaviours.



Training iteration

This does not establish human-like thought or guarantee truth.

Sources: DeepSeek-V3 Technical Report · Nature (R1)

R1-Zero, R1 and distillation

Distinguish the RL experiment, the practical training pipeline and capability transfer.

01

R1-Zero

Extensive RL without initial SFT produces useful behaviours and readability problems.

02

R1

Cold-start, supervised and RL stages improve utility and clarity.

03

Distillation

Smaller models learn from traces or outputs of larger systems.

04

Caution

Evaluate length, cost, errors, bias and safety for each task.

Nature published the DeepSeek-R1 study in 2025.

Sources: DeepSeek-V3 Technical Report · Nature (R1)

BLOCK 04 / 07

Qwen: a family of architectures

Open models and multimodal capabilities

What changes between Qwen3 and Qwen3.5?

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Qwen3: deliberation or a direct answer

Hybrid modes help manage reasoning effort and latency.

01

NON-THINKING

Direct responses for simple requests, conversation and tasks with low complexity.

02

THINKING

More deliberative processing for reasoning, code and decision sequences.

Choose the mode according to difficulty, budget and reliability.

Sources: Alibaba Group · Qwen Team, 2025–2026

Qwen3.5: multimodal architecture

The 2026 announcement combines hybrid attention with sparse experts.

397 B

Total parameters in the announced model

17 B

Parameters activated per forward pass

These figures do not establish equal quality, cost or performance across families.

Qwen3.5-397B-A17B. Figures reported by Alibaba in February 2026.

Sources: Alibaba Group · Qwen Team, 2025–2026

Why hybrid attention?

Architectures balance context, memory and performance.

01

Standard attention

Rich interactions among tokens, with costs that grow with context.

02

Linear attention

Reduces selected costs using recurrent states or alternative formulations.

03

Combination

Interleaves blocks to retain capabilities and improve efficiency.

04

Conditions

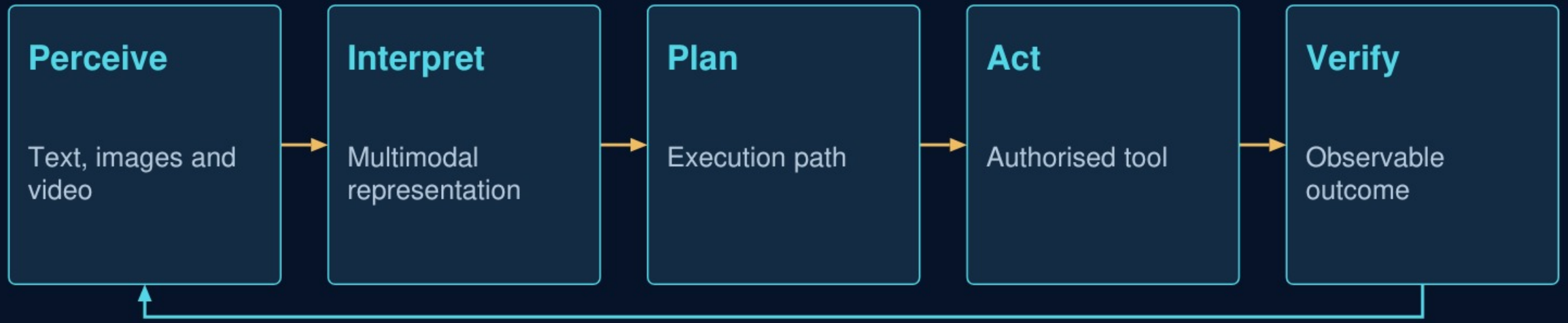
Results depend on hardware, sequence length and workload.

Performance advantages depend on the task and execution conditions.

Sources: Alibaba Group · Qwen Team, 2025–2026

Native agents and multimodal perception

Text and images support decisions and tool use.



Agentic systems require orchestration and boundaries as well as multimodal models.

Sources: Alibaba Group · Qwen Team, 2025–2026

DeepSeek and Qwen: an academic comparison

Compare documented design objectives and evidence.

A

DEEPSEEK

Architectural efficiency with MoE and MLA, and reinforcement learning for reasoning, particularly in V3 and R1.

B

QWEN

A broad family with hybrid reasoning modes, dense and MoE variants, and multimodal models for agentic applications.

A valid comparison needs the same date, benchmark, hardware and protocol.

Sources: Alibaba Group · Qwen Team, 2025–2026

BLOCK 05 / 07

Reasoning, optimisation and cost

From probability to verification

When is more reasoning worthwhile, and when is a tool more appropriate?

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Training and inference

Training expenditure does not describe the marginal cost of a query.

01

TRAINING

Weight updates using large datasets, optimisation, infrastructure and experimentation.

02

INFERENCE

Token generation incurs computing, memory, latency, concurrency and output costs.

A low query price can coexist with a high cost per completed task.

Sources: Nature R1 · Qwen3 / Qwen3.5 · Stanford AI Index

Reasoning budgets

More deliberation tokens can help difficult tasks while increasing cost.

01

Quality

Success validated against independent criteria.

02

Latency

Time to an accepted solution, including completion.

03

Consumption

Tokens, GPU work, memory, searches and retries.

04

Risk

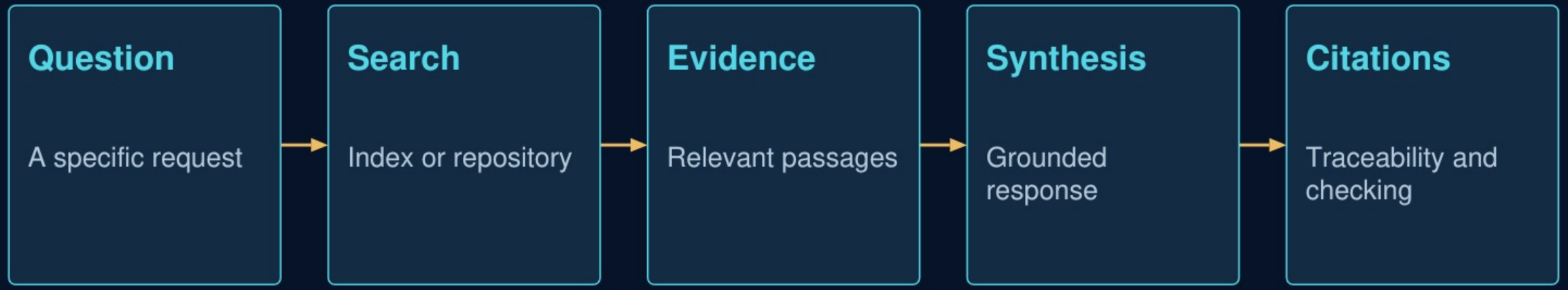
Prolonged errors, loops and unwarranted confidence.

The simulator adjusts difficulty and reasoning budgets in a conceptual model.

Sources: Nature R1 · Qwen3 / Qwen3.5 · Stanford AI Index

RAG: retrieval before generation

Retrieval connects generation with documents and traceable sources.



Retrieved documents may contain malicious instructions.

Sources: Nature R1 · Qwen3 / Qwen3.5 · Stanford AI Index

Fine-tuning, RAG and tools

Different approaches address different problems.

01

Fine-tuning

Adapt style, format and task patterns.

02

RAG

Retrieve changing, traceable knowledge.

03

Tool use

Perform operations and inspect external state.

04

Rules

Enforce constraints when an exact policy is preferable.

Choose according to update needs, safety and cost.

Sources: Nature R1 · Qwen3 / Qwen3.5 · Stanford AI Index

The economics of inference

Optimise cost per correctly completed task alongside token prices.

01

Effective cost

Model, tools, verification and retries.

02

Throughput

Correct tasks per unit of time or budget.

03

Scaling

Concurrency, caching, quantisation and batching.

04

Governance

Privacy, auditing, licences and processes.

Architecture choices also affect how organisations work.

Sources: Nature R1 · Qwen3 / Qwen3.5 · Stanford AI Index

BLOCK 06 / 07

Agents and multi-agent systems

From answering questions to executing workflows

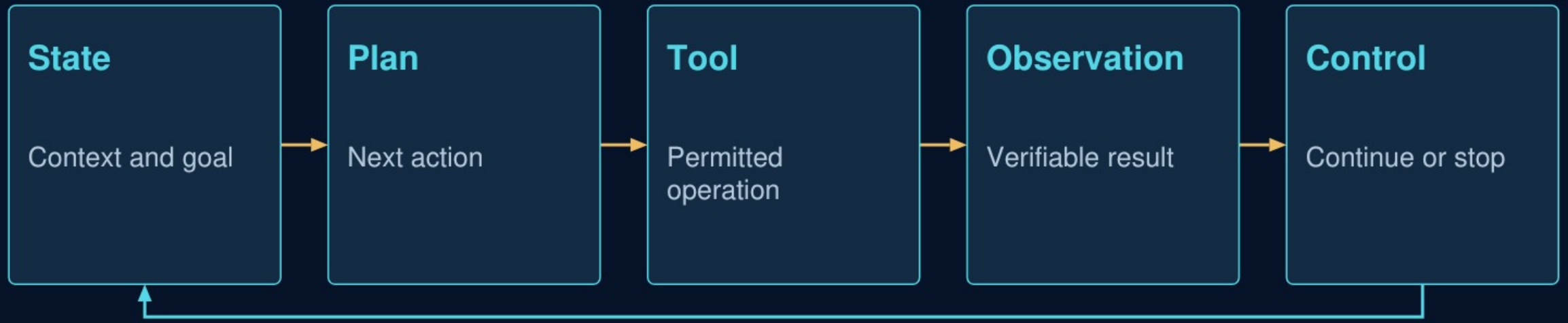
How does a model become part of an operational architecture?

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An AI agent

A loop that observes, decides, uses tools and checks state.



Continue with a new observation

Autonomy still requires supervision.

Sources: Hugging Face Agents Course · Stanford · evaluation studies

Coordination architectures

Specialisation introduces overhead and additional failure modes.

01

Single agent

Simple to operate and fast for bounded tasks.

02

Sequential

A chain of specialised stages with explicit handoffs.

03

Parallel

Independent tasks executed simultaneously.

04

Hierarchical

A supervisor allocates and validates work.

Use the simplest architecture that meets the task requirements.

Sources: Hugging Face Agents Course · Stanford · evaluation studies

A research agent

Reproducible research needs a clear workflow and traceability.

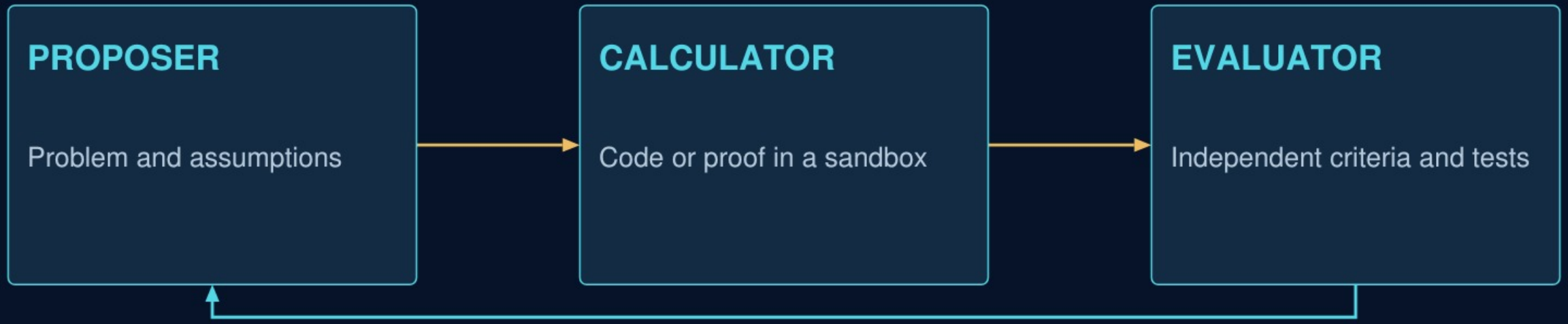


Measure fabricated citations as an explicit failure mode.

Sources: Hugging Face Agents Course · Stanford · evaluation studies

A mathematical agent and evaluator

Separating proposals, calculations and verification exposes hidden errors.



On failure: revise the proposal

SUPERVISOR: resource and retry limits

A model evaluator may share the same biases. Include deterministic tests.

Sources: Hugging Face Agents Course · Stanford · evaluation studies

MCP, tools and permissions

Structured interfaces support connectivity while risks remain.

01

Schema

Typed inputs and understandable outputs.

02

Permissions

Least privilege and approvals for sensitive actions.

03

State

Logs of each action and result.

04

Sandbox

Isolation, resource limits and reproducibility.

Retrieved text must retain its lower trust level.

Sources: Hugging Face Agents Course · Stanford · evaluation studies

BLOCK 07 / 07

Evaluation, safety and deployment

From demonstrations to reliable systems

How can capability, cost and safety be tested with distinct metrics?

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Benchmarks: scope and limitations

Benchmark improvements do not guarantee performance on real tasks.

01

Capability

Task sets, criteria and protocols.

02

Contamination

The risk that training included test examples.

03

Generalisation

New tests and changing distributions.

04

Final metrics

Accuracy, success rate, robustness and cost.

Record the version, date and execution conditions.

Sources: Stanford AI Index 2026 · GAIA · OSWorld · OWASP

Evaluating agents means testing trajectories

Measure the process and execution as well as the final answer.



GAIA and OSWorld include tasks involving tools.

Sources: Stanford AI Index 2026 · GAIA · OSWorld · OWASP

Safety: injection and actions

Hostile instructions can arrive inside documents or tool results.

01

Prompt injection

Lower-trust sources try to redirect the plan.

02

Exfiltration

Sensitive data sent to unauthorised destinations.

03

Tools

Actions with excessive permissions.

04

Defences

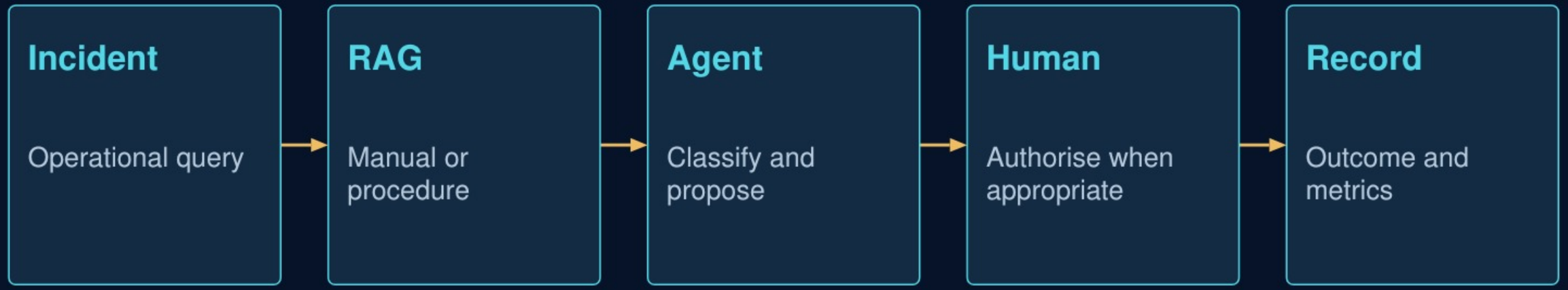
Isolation, validation, logs and approval.

Text filtering alone does not guarantee protection.

Sources: Stanford AI Index 2026 · GAIA · OSWorld · OWASP

An industrial support example

Combine agents, sources, rules, human review and measurable returns.



This is a conceptual design, not an implemented system.

Sources: Stanford AI Index 2026 · GAIA · OSWorld · OWASP

Designing responsible adoption

A sound programme connects technical performance with institutional economics.

01

Pilot

Narrow scope and representative data.

02

Evaluation

Compare against a human baseline or a simple rule.

03

Governance

Privacy, rights and permissions.

04

Scale

Capacity, support and continuous monitoring.

Which processes actually change?

Sources: Stanford AI Index 2026 · GAIA · OSWorld · OWASP

CHINA AND THE AI FRONTIER

The frontier is a complete system.

Architecture · data · hardware · agents · evaluation · institutions

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